Monroe County: Climate Health

Temperature Days $\geq 88°F$

County Historical Climate Data: May, June, and August of Year 1994 - 2024

Introduction: Monroe County's vulnerable populations, includes significant percentages of children, the elderly, individuals with chronic illnesses, high rates of ER visits and hospital readmission, at risk of heat-related health impacts(Click: RHIO, 2021). Addressing these risks requires a localized framework that utilizes daily maximum temperatures to guide effective risk mitigation strategies. Historical climate data plays a crucial role in optimizing cooling center availability and resource allocation by identifying patterns of consecutive days with temperatures exceeding critical thresholds. These patterns, analyzed across various dates, months, and years, provide valuable insights for effective cooling center management within the County. Furthermore, studies from 2010[2], 2011[1], and 2017[3] reveal that the threshold for triggering heat-related illnesses varies significantly, influenced by factors such as humidity (wet bulb temperature) and individual susceptibility. This underscores the importance of localized definitions and targeted interventions to enhance heatwave management efforts.

Heatwaves are typically defined as periods lasting 2–3 days or more where temperatures exceed the usual threshold for a specific region, often compounded by high humidity levels. These conditions significantly heighten the risk of heat-related illnesses, such as heat exhaustion and heatstroke. Regulatory standards frequently consider a heatwave as temperatures of 96°F (36°C) or higher sustained over 2–3 consecutive days. However, peer-reviewed studies[1,2,3,4] highlight that heat-related illnesses can be triggered at lower temperatures depending on individual vulnerabilities, such as age, preexisting health conditions, and local environmental factors. Ambient conditions, including inadequate vegetation, urban heat islands, and poorly ventilated housing, further amplify the effects of heatwaves.

Summary of Key Findings

Summary Across Year 1994-2024: An analysis of heatwave data from 1994–2024 reveals that days with temperatures of 88°F and above, often accompanied by high relative humidity (RH), exacerbate the perceived temperature and its physiological impact. Such days are most frequent in July, accounting for 45.26% of occurrences, with an average of 4.93 days per month. August follows with 25.08% and an average of 2.73 days per month, while June and May contribute 22.94% and 6.73%4, respectively. Over the 30-year period, there have been 327 days exceeding 88°F, with an overall yearly average of 10.9 days and a 30-year monthly average of 2.73 days. July is underscores as the peak period for high-temperature events. [a]

[a] Health and Heat Resources: visit link

Top Priority Week for Heatwave Preparedness Based on Climate Historical Data

Based on the frequency of Date with heat(88°F and above) and the consecutive days:

- **Top Priority Week: Mid-Month (Dates 14–21):** This period should be prioritized as it coincides with several consecutive hot days. Specifically, notable peaks occur on Dates 14, 16, and 17—each appearing 13–15 times.
- **Secondary Priority Week: Fourth Week (Dates 27–31):** High temperatures are also frequent during the last week of the month, particularly on Dates 27, 28, and 31. Preparation and flexibility are essential during this week.

Most Heatwave Days (To aid early allocation of cooling center resources and staff scheduling):

- **Top Priority Days:** - Dates 4 and 17: **15 occurrences** (4.6%). - Date 6: **14 occurrences** (4.29%).
- **High Priority Days:** - Dates 14, 16, 20, and 31: Tied with **13 occurrences** (3.99%).
- **Moderate Priority Days:** - Dates 18, 19, and 9: Tied with **12 occurrences** (3.68%).
- **Additional Preparation Days:** - Dates 1, 21, 27, 28, and 29: **11 occurrences each** (3.37%).

Summary Table of May–August Temperature $\geq 88°F$ (1994–2024)

Year	No. of Days $\geq 88°F$	Date Observed
1994	10	15, 16, 17, 18, 4, 6, 20, 21, 4, 20
1995	13	18, 19, 13, 14, 15, 1, 3, 14, 15, 19, 20, 21, 31
1996	3	19, 5, 6
1997	2	14, 15
1998	2	16, 25
1999	18	6, 7, 12, 27, 3, 4, 5, 6, 15, 16, 17, 18, 23, 24, 27, 28, 30, 31
2000	1	14
2001	13	14, 15, 19, 29, 23, 24, 1, 2, 5, 6, 7, 8, 9
2002	22	21, 23, 25, 1, 2, 4, 8, 17, 21, 22, 28, 29, 31, 1, 2, 4, 11, 12, 13, 14, 15, 16
2003	3	25, 26, 4
2004	1	9
2005	16	9, 24, 25, 27, 4, 11, 12, 13, 14, 18, 25, 26, 3, 4, 9, 10
2006	13	29, 30, 17, 18, 14, 16, 17, 20, 26, 31, 1, 2, 18
2007	16	24, 25, 31, 8, 12, 26, 27, 9, 10, 31, 1, 2, 3, 24, 25, 29
2008	11	6, 8, 9, 13, 7, 8, 12, 16, 17, 18, 19
2009	1	17
2010	9	31, 5, 6, 7, 8, 17, 3, 30, 31
2011	12	31, 8, 10, 11, 12, 16, 17, 18, 20, 21, 22, 23
2012	19	28, 11, 20, 21, 29, 30, 4, 6, 12, 13, 14, 16, 17, 23, 31, 3, 4, 25, 31
2013	10	30, 31, 1, 23, 14, 15, 16, 17, 18, 19
2014	5	17, 28, 30, 1, 22
2015	7	8, 9, 19, 27, 28, 29, 17
2016	28	27, 28, 19, 20, 26, 27, 5, 6, 7, 12, 13, 14, 21, 22, 23, 27, 3, 4, 5, 9, 10, 11, 12, 13, 16, 20, 24, 28
2017	2	12, 18
2018	22	30, 17, 18, 29, 30, 1, 2, 3, 4, 5, 9, 10, 13, 14, 15, 16, 20, 4, 5, 6, 28, 29
2019	8	28, 4, 10, 16, 19, 20, 27, 29
2020	13	26, 21, 22, 2, 5, 6, 7, 8, 9, 10, 19, 10, 24
2021	14	21, 6, 7, 26, 27, 28, 29, 5, 13, 9, 11, 25, 26, 28
2022	17	31, 16, 21, 22, 30, 1, 11, 19, 20, 22, 23, 24, 3, 6, 8, 20, 29
2023	2	2, 6
2024	13	21, 17, 18, 19, 20, 3, 5, 9, 14, 30, 1, 4, 27

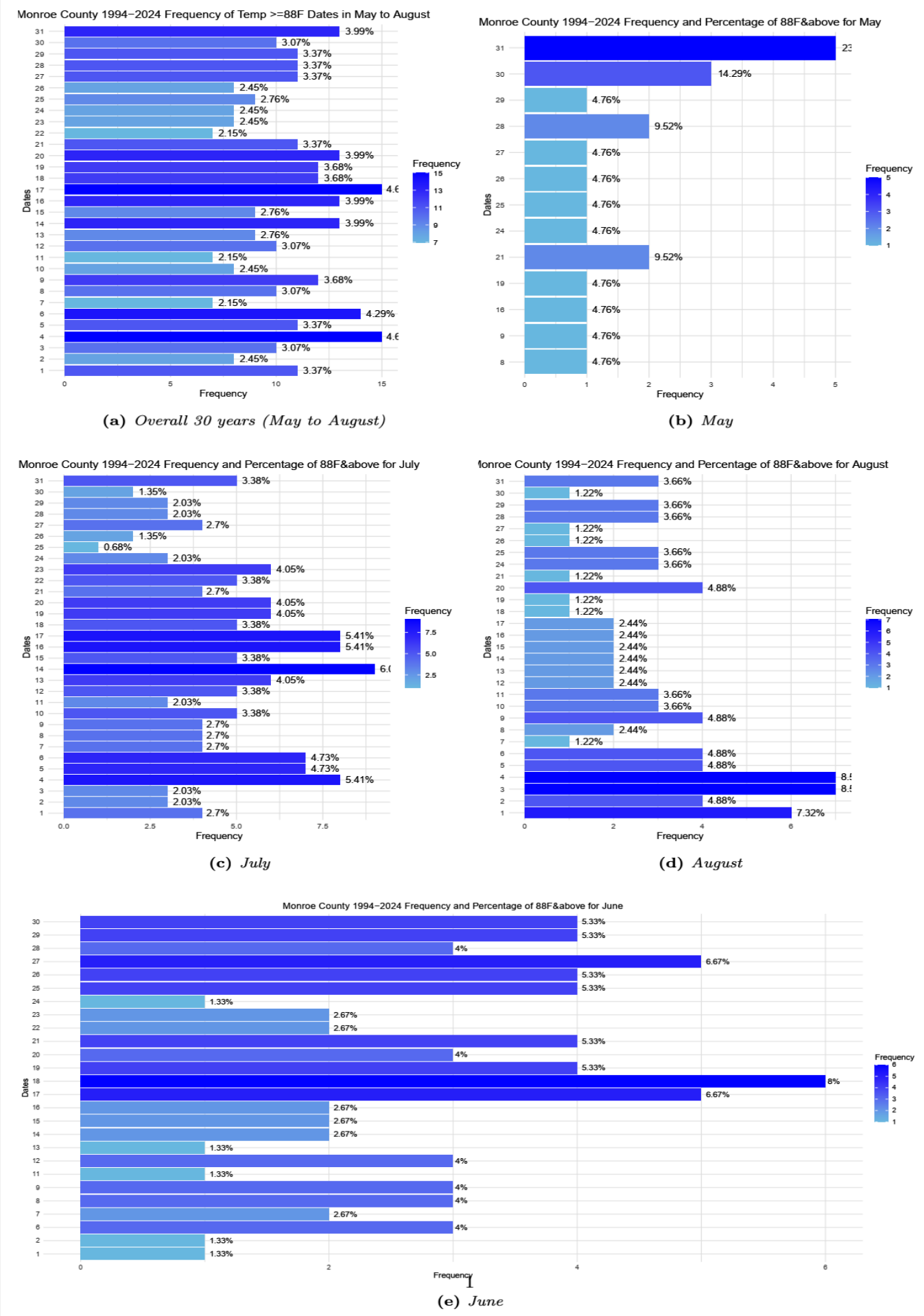
June Heatwave Patterns from 1994-2024

- **Top Week:**
  - The week of **June 17–21** shows the highest heatwave activity, with several years recording multiple heatwave days and several consecutive days.
- **Overall Heatwave Trends:**
  - **June 25–30** frequently shows clusters of heatwave days, This indicates a need to prepare for late June heatwave clusters as well. **June 6,8-9** is another key period, with multiple consecutive heatwave days.

August Heatwave Patterns from 1994-2024

- **Top Dates: August 3-4** stands out as the most frequent period, appearing in multiple years.
- **Top Week:** The week of **August 1-9** shows several consecutive heatwave days. **Highlighting early-month staff and resource deployment to cooling centers.**
- **Late-August Heatwaves: August 24-31** also shows notable consecutive heatwave appearing in multiple years.

Exploratory Data Analysis



(a) *Overall 30 years (May to August)*

(b) *May*

(c) *July*

(d) *August*

(e) *June*

Figure 1: Historical Data 1994-2024, May to August

July Most Frequent and Consecutive Days with Higher Percentage from 1994-2024

High Priority Days by Frequency:

- **Date 14:** The highest frequency of **6 occurrences (6.0%)**.
- **Dates 16-17:** Both with **5.41%**, indicating multiple consecutive extreme heat days and **date 4**.
- **Dates 5 and 6:** Both with **4.73%**, showing frequent early-month heatwaves.
- **Dates 19, 20, and 23:** All tied with **4.05%**.

Days with Consecutiveness:

- **Mid-Month Cluster (14–17):** Several high-frequency days occur consecutively, making this period critical for prioritization.
- **End of Month Cluster (27–31):** Days like **27, 28, and 31** show consistent heat, albeit with slightly lower percentages.

Recommendations

- Based on the identified patterns of consecutive days with temperatures exceeding 88°F—indicative of heatwaves often compounded by high relative humidity:
  - It is recommended that cooling centers openings and Staff deployment to the centers within the county remain fully operational for at least 10–12 days of month from May through August.
  - Compare current cooling centers capacities with past hot season visitor records or estimates to anticipate and address rising needs, as hot days are projected to increase.
  - Operational days and resources should be strategically allocated to align with the peak heat periods. Specifically, 45% of the total operational days should be prioritized for July, followed by 25% in August, and 23% in June, with May requiring fewer days.

Monroe County Health Indicators:

This table consolidates key demographic and health data, emphasizing the need for a framework to mitigate heat-related risks among vulnerable populations.

Indicator	Value	Comparison
Children Aged 0-4	39,942 (5.28%)	
Elderly Aged 65+	135,497 (17.91%)	
Children in Poverty	29,116 (19.11%)	Below Federal Poverty Level (FPL)
Coronary Heart Disease Deaths	3,996 (107.1 per 100,000)	Above state & national averages
Stroke Deaths	1,544 (41.4 per 100,000)	Above state & national averages
ER Visit Rate	724.6 per 1,000	Higher than state rate (560.7)
Medicare 30-day Readmission	21.5%	Above New York (18.5%) & U.S. average (17.8%)

Health Indicators extracted from RocHealthData, 2021

[1] The month of July stands out as the critical period with the highest heat frequency.

NOTE: Historical data exploration offers limited predictive power for future heat events. However, it provides a baseline for planning. The minimum average number of expected hot days is likely to rise in the future, emphasizing the need for proactive measures.

Reference:

1. Anderson, G. B., & Bell, M. L. (2011). Heat waves in the United States: Mortality risk during heat waves and effect modification by heat wave characteristics in 43 U.S. communities. *Environmental Health Perspectives, 119*(2), 210-218. https://ehp.niehs.nih.gov/doi/pdf/10.1289/ehp.1002313

2. Hajat, S., O'Connor, M., & Kosatsky, T. (2010). Health effects of hot weather: From awareness of risk factors to effective health protection. *The Lancet, 375*(9717), 856-863. https://pubmed.ncbi.nlm.nih.gov/20153519/

3. Mora, C., Dousset, B., Caldwell, I. R., et al. (2017). Global risk of deadly heat. *Nature Climate Change, 7*(7), 501-506. https://www.nature.com/articles/nclimate3322

4. Data Source from Roc Health Data: Monroe County Local Health Assessment (HealthData Indicator Report). https://rochealthdata.org/

5. Weather Underground. (2025). *Monthly weather history for Rochester, NY: 2025.* Retrieved from https://www.wunderground.com/history/monthly/us/ny/rochester/KROC/date/2015-12